

Macro Narrative Dynamics

A Quantitative Framework for Measuring Narrative Evolution in U.S. Financial Discourse

Project Plan and Architecture Document

April 30, 2026

Document Purpose

This document is the operational backbone of the project. It consolidates every decision, methodology choice, risk mitigation, and architectural commitment we have made through extensive design discussion. It is intended to remain authoritative throughout the project’s execution. Footnoted items marked **[MVP]** can be cut for a fastest-possible build; items marked **[Compress]** indicate compressible scope without losing the project’s core contribution.

Contents

1	Executive Summary	4
2	Background and Motivation	4
2.1	The Problem Space	4
2.2	What This Project Does and Does Not Do	5
2.3	Why This Matters	5
3	Scope and Boundaries	5
3.1	Domain Scope	5
3.2	Geographic and Linguistic Scope	5
3.3	Temporal Scope	6
4	Theoretical Framework	6
4.1	Narratives as Epidemics	6
4.2	Why This Framework Is Defensible	7
4.3	Multi-Model Robustness	7
5	System Architecture	7
5.1	Pipeline Overview	7
5.2	Stage 1: Data Ingestion	8
5.3	Stage 2: Filtering and Deduplication	8
5.4	Stage 3: Embedding	8
5.5	Stage 4: Clustering	9
5.6	Stage 5: Dynamics Fitting	9
5.7	Stage 6: Stage Classification	9

5.8	Stage 7: Dashboard	9
6	Data Sources	9
6.1	Primary Text Sources	10
6.2	Validation Data Sources	10
7	Embedding Strategy and Look-Ahead Bias	10
7.1	The Concern	11
7.2	Our Approach	11
7.3	Why This Does Not Undermine the Project	11
8	Methodological Rigor and Risk Mitigations	11
8.1	Narrative Identity Risk	12
8.2	Cluster Stability Risk	12
8.3	Reverse Causality Risk	12
8.4	Multiple-Testing and False Discovery Risk	13
8.5	Survivorship Bias Risk	13
8.6	Data Quality Risk	13
8.7	Reproducibility Risk	14
9	Kill Criteria	14
9.1	Fallback Scope (Insurance Policy)	14
10	Validation Strategy	14
10.1	Anchor Narratives	14
10.2	Fizzled Counterparts	15
10.3	Out-of-Sample Discipline	15
10.4	Statistical Reporting Standards	15
11	Dashboard Design	15
11.1	Two Primary Views	15
11.2	Empty-State Handling	16
11.3	Onboarding	16
11.4	Live Updating Architecture	16
12	Build Plan	16
12.1	Phase 1: Pilot and De-Risking	16
12.2	Phase 2: Data Pipeline	17
12.3	Phase 3: Embedding, Clustering, and Dynamics	17
12.4	Phase 4: Validation	17
12.5	Phase 5: Dashboard Build	17
12.6	Phase 6: Live Updating	18
12.7	Phase 7: Writeup and Reproducibility	18
12.8	Speedrun MVP Summary	18
13	Deliverables	18
13.1	Primary Deliverables	18
13.2	Secondary Deliverables	19

14 Resume and Application Signal	19
A Pre-Specified Parameters	19
B Glossary	20

1 Executive Summary

This project builds a quantitative measurement framework for tracking how macro-financial narratives form, propagate, peak, and decay in U.S. financial discourse from 2010 to the present. The system clusters published financial articles into coherent narratives using transformer-based semantic embeddings, fits epidemiological growth models to each narrative’s life-cycle, and surfaces this analysis through a public web dashboard that updates weekly.

The project’s core intellectual contribution is the operationalization of Robert Shiller’s narrative economics framework with modern natural language processing tooling, applied with rigorous methodology to U.S. financial discourse. The framework brings quantitative measurement to a domain that has remained largely qualitative since Shiller’s foundational work, and produces a publicly-deployed tool that allows users to observe narrative dynamics descriptively rather than predictively.

The project is deliberately scoped as a *descriptive* and *historical* measurement system, not a predictive trading tool. This reframe is methodologically important: it sidesteps causal-inference and reverse-causality concerns that would undermine predictive claims, and it produces a tool whose value is in helping users *understand* how financial discourse evolves rather than betting on its consequences.

One-Paragraph Pitch

Macro narratives in U.S. financial discourse—about inflation, recession, monetary policy, and the macro implications of major shocks—spread through analytical writing in patterns resembling epidemics. We measure these dynamics by clustering articles into coherent narratives, fitting epidemiological models to their growth curves, and tracking their life-cycle stages. The system updates weekly with the latest narrative state of U.S. financial discourse and surfaces narratives currently emerging that resemble historical predecessors. The contribution is a measurement framework that brings quantitative rigor to a domain—narrative economics—that has been mostly qualitative since Shiller’s foundational work.

2 Background and Motivation

2.1 The Problem Space

Financial markets are not driven solely by fundamentals. Stories, framings, and narratives shape how participants interpret information, form expectations, and make decisions. Robert Shiller’s *Narrative Economics* (Shiller, 2017, 2019) formalized this intuition: narratives spread through populations like epidemics, with measurable transmission rates and life-cycles, and significant economic events often coincide with narrative phenomena. Examples include the “transitory inflation” debate of 2021–2022, the “soft landing versus hard landing” framing of 2023–2024, and the “is this 2008 again” framing during the March 2023 banking stress.

While this framework is intellectually compelling and widely cited, its empirical operationalization remains underdeveloped. Existing work in this space falls into two camps:

1. **Qualitative narrative analysis** (the bulk of Shiller’s own empirical work) which uses keyword frequencies in Google Ngrams or hand-curated narrative themes. This is interpretable but lacks the granularity that modern natural language processing makes possible.

2. **Quantitative text-based finance** (large literature) which uses sentiment scores, topic models, and embeddings to predict market outcomes. This is methodologically sophisticated but typically does not engage with narratives as dynamic, propagating entities with life-cycles.

The gap is a system that combines modern NLP’s semantic granularity with epidemiological modeling of narrative dynamics, applied with rigorous methodology and presented as a measurement tool rather than a prediction engine.

2.2 What This Project Does and Does Not Do

Scope of Claims

This project DOES: measure narrative dynamics in U.S. financial discourse using transformer embeddings and epidemiological models; deploy a public dashboard for descriptive observation of these dynamics; validate methodology against documented historical narratives; report exploratory associations with market outcomes as a footnote.

This project DOES NOT: claim causal influence of narratives on market outcomes; produce trading signals; predict market movements; cover non-English or non-U.S. discourse; analyze short-form social media content; engage in mechanistic interpretability or model architecture work.

2.3 Why This Matters

The system’s value is twofold. First, as a research artifact, it advances the empirical study of narrative economics by providing the first systematic, reproducible, publicly-deployed measurement framework for narrative dynamics in financial discourse. Second, as a public tool, it gives any visitor—specialist or not—a structured window into how U.S. financial discourse evolves: what is being discussed, how those discussions are growing, and what historical narratives they resemble.

3 Scope and Boundaries

3.1 Domain Scope

The project covers **macro-financial narrative discourse** as it appears in U.S. financial publications. Specifically:

- **Included topical areas:** inflation expectations and cost-of-living narratives; recession and business-cycle narratives; monetary policy framings; major macro shocks and their interpretive aftermath (banking stress, supply disruptions, geopolitical events with macro-financial dimensions).
- **Excluded topical areas:** single-stock analyst commentary; sector-specific narratives without macro implications; pure technical-trading commentary; equity micro-stories; cryptocurrency speculation outside macro-relevant framings.

3.2 Geographic and Linguistic Scope

The project covers **U.S. financial press in English**, including U.S. press coverage of foreign markets, foreign policies, and foreign financial events. The defining criterion is the publication

source, not the subject matter: a Wall Street Journal article about European Central Bank policy is in scope; a French-language Le Monde article about the same event is not.

This scope is acknowledged explicitly as a limitation: findings reflect Anglo-American financial discourse, not global narrative dynamics. Extension to other discourse communities (Eurozone, East Asian, emerging market) is identified as future work.

3.3 Temporal Scope

Historical analysis: 2010 through the present. This window captures the post-financial-crisis era with cleaner data and continuous web archives. It includes multiple regime shifts (zero-interest-rate era, taper tantrum of 2013, COVID shock of 2020, inflation regime of 2022, banking stress of 2023, AI-capex boom of 2023–present), providing rich material for narrative analysis.

Live extension: weekly updates going forward. The system extends its analysis with new articles each week, refitting parameters on changed clusters and refreshing the dashboard.

On Pre-2010 Data

We deliberately do not include pre-2010 data. GDELT 2.0 (the better-structured version) only goes back to 2015; GDELT 1.0 covers earlier years but with messier metadata. Pre-2008 data has additional regime-change complications because the modern financial-news ecosystem and the post-crisis monetary regime represent structural breaks. 2010 is a clean compromise: post-crisis, well-archived, pre-dating most current narratives.

4 Theoretical Framework

4.1 Narratives as Epidemics

The core analytical move is to model narratives as epidemic phenomena. Following the SIR (Susceptible–Infected–Recovered) framework (Kermack and McKendrick, 1927), we treat each narrative as an infectious idea spreading through a discourse community. At any time t , the population of potential narrative-carriers (here, articles or pieces of analytical writing) is partitioned into three compartments:

$$\frac{dS}{dt} = -\beta \cdot \frac{S \cdot I}{N} \quad (1)$$

$$\frac{dI}{dt} = \beta \cdot \frac{S \cdot I}{N} - \gamma \cdot I \quad (2)$$

$$\frac{dR}{dt} = \gamma \cdot I \quad (3)$$

where $S(t)$ represents the susceptible discourse capacity, $I(t)$ represents currently-active narrative-carrying articles, $R(t)$ represents recovered or stale articles no longer actively propagating the narrative, β is the transmission rate, γ is the recovery rate, and N is the total discourse population.

The basic reproduction number $R_0 = \beta/\gamma$ characterizes a narrative’s transmissibility. Narratives with $R_0 > 1$ propagate; those with $R_0 < 1$ die out. The dynamics produce characteristic life-cycle patterns: an emergence phase, exponential growth, peak attention, and decay, all of which are observable in article-volume time series.

4.2 Why This Framework Is Defensible

The SIR analogy is not merely metaphorical. Kermack and McKendrick’s epidemiological framework has been validated in non-biological contexts, including news propagation (Vitanov et al., 2024), social media information spread, and idea diffusion. Shiller (2017) explicitly invokes the epidemic analogy:

“We need to incorporate the contagion of narratives into economic theory. Otherwise, we remain blind to a very real, very palpable, very important mechanism for economic change.”

Our project operationalizes this analogy with quantitative rigor.

4.3 Multi-Model Robustness

We do not commit exclusively to SIR. Each narrative’s growth curve is fit with multiple parametric models in parallel, allowing cross-model validation and graceful degradation when one model fits poorly:

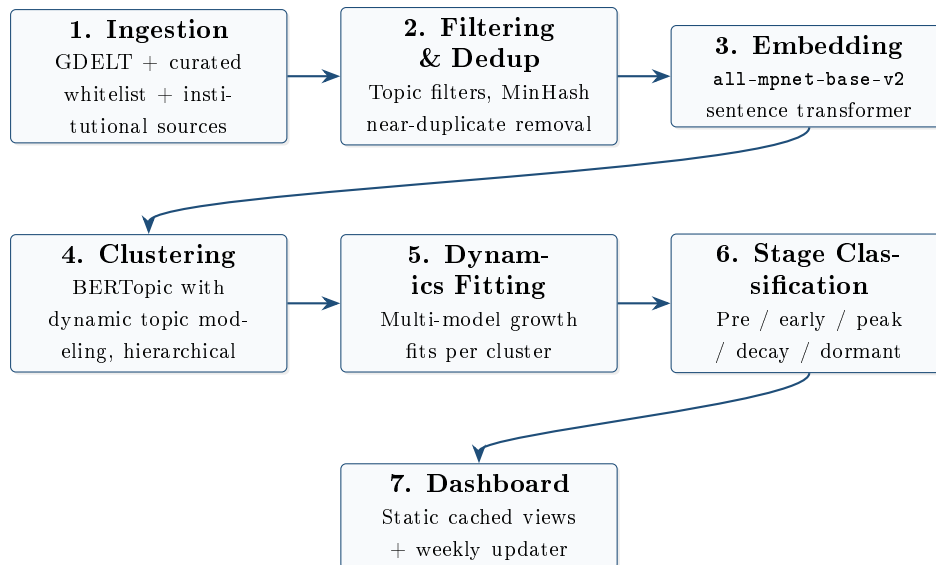
- **SIR / SEIR**: epidemiologically motivated, three-to-four parameters.
- **Logistic growth**: $f(t) = L / (1 + e^{-k(t-t_0)})$, two parameters, robust to noise.
- **Gompertz growth**: $f(t) = L \cdot e^{-e^{-k(t-t_0)}}$, asymmetric peak, common in diffusion modeling.
- **Simple exponential**: baseline for very early-stage narratives.

If SIR fits poorly across the validation set, dashboard outputs and analysis fall back to logistic growth or to non-parametric raw curve features (peak velocity, time-to-peak, decay slope).

5 System Architecture

5.1 Pipeline Overview

The system is organized as a pipeline of seven stages, each producing a checkpointed artifact that downstream stages consume. This modular structure ensures that partial completion still produces useful outputs, and that failures or pivots in one stage do not require redoing upstream work.



5.2 Stage 1: Data Ingestion

The ingestion layer pulls article-level data from three source tiers. See Section 6 for full source specifications.

Tier A: GDELT 2.0 as discovery layer. GDELT provides URL-and-headline metadata for articles from a wide range of outlets. We use it to discover articles from our curated whitelist (Tier B), not as a primary text source.

Tier B: Curated U.S. financial press whitelist. A pre-specified list of approximately 30–50 U.S. financial publications (see Section 6). Full text accessed through institutional library databases (Factiva, ProQuest News, LexisNexis Academic) where available.

Tier C: Institutional sources. Federal Reserve communications, IMF and BIS publications, NBER working papers, OECD outlook documents. All free and machine-readable.

5.3 Stage 2: Filtering and Deduplication

After ingestion, two cleaning operations:

Topic filtering. A hybrid filter combining keyword lists for macro-financial relevance with embedding-based similarity to a small seed set of paradigmatically macro articles. Articles failing either check are excluded.

Near-duplicate removal. MinHash-based near-duplicate detection within rolling 48-hour windows removes redistributions, syndicated content, and minor edits of the same article. We retain one canonical version per duplicate cluster.

5.4 Stage 3: Embedding

Each retained article (typically headline plus first 500–800 tokens of body, to control compute) is encoded using the `all-mpnet-base-v2` sentence transformer. The choice of a general-purpose rather than finance-specific embedding model is deliberate; see Section 7 for the rationale concerning look-ahead bias.

5.5 Stage 4: Clustering

Clustering uses BERTopic (Grootendorst, 2022) with dynamic topic modeling extensions:

- UMAP dimensionality reduction (pre-specified parameters; see Appendix A).
- HDBSCAN density-based clustering producing variable-sized clusters.
- Class-based TF-IDF for cluster representation, allowing time-resolved keyword tracking.
- Hierarchical merging into three granularity levels: fine (200+ clusters), medium (40–80), coarse (10–20).

A narrative is operationally defined as a cluster output by this pipeline at the medium granularity level. Fine and coarse levels are retained for hierarchical exploration in the dashboard.

5.6 Stage 5: Dynamics Fitting

For each narrative cluster, we extract the time series of articles per day assigned to that cluster, smooth it with a 7-day centered moving average to remove weekend dips, and fit four parametric growth models in parallel: SIR, logistic, Gompertz, and exponential. Fitting uses Bayesian inference with weakly-informative priors (PyMC implementation), producing full posterior distributions over parameters.

The Bayesian approach yields not only point estimates but explicit credible intervals on R_0 , peak time, and decay rate—essential for honest reporting of uncertainty.

5.7 Stage 6: Stage Classification

Each narrative is classified into a life-cycle stage based on its fitted curve:

- **Pre-emergence:** fewer than N_{thresh} articles, R_0 poorly identified.
- **Early-spread:** $R_0 > 1$, currently in growth phase, has not reached peak.
- **Peak:** at or near maximum daily article volume, growth rate near zero.
- **Decay:** post-peak, declining article volume.
- **Dormant:** very low volume, $R_0 < 1$, narrative effectively concluded.

Threshold values are pre-specified before any analysis (see Appendix A).

5.8 Stage 7: Dashboard

The dashboard is a public web tool with two views (see Section 11). It reads pre-computed static artifacts—no live computation occurs at user request time. Updates occur via a weekly batch job that extends the historical analysis with new articles, refits parameters on changed clusters, and refreshes the static artifacts.

6 Data Sources

Source	Provides	Access	License
GDELT 2.0	Article URLs, headlines, source domains, dates from global news outlets	Free, BigQuery + <code>gdeltpyR</code>	Open, attribution
Curated U.S. press	Full text from ~30–50 outlets including WSJ, Bloomberg, FT (US), Reuters, Barron’s, Economist, Foreign Affairs	Library databases (Factiva, ProQuest)	Academic use
Federal Reserve	FOMC statements, minutes, Beige Book, speeches, MPRs, research papers	<code>federalreserve.gov</code>	Public domain
IMF / BIS / OECD	Working papers, country reports, outlook documents, monetary policy reports	Institutional websites	Open academic
NBER + SSRN	Academic finance and macro working papers, conference papers	Institutional sites	Free academic

Table 1: Primary text sources for narrative analysis.

6.1 Primary Text Sources

Critical pre-flight check. Before week-one work begins, confirm UChicago library access to Factiva or ProQuest News for historical full-text access to U.S. financial press. Run a test query for representative outlets (WSJ, FT, Bloomberg) and verify programmatic retrieval is feasible under academic-use terms. This determines whether the project’s primary text layer is operationally available.

6.2 Validation Data Sources

Source	Provides	Access
FRED	CPI, PCE, unemployment, yield curve, breakeven inflation, sector indexes, VIX, credit spreads	<code>fredapi</code> , free
Yahoo / Alpha Vantage	Daily prices and volumes for individual securities and ETFs	<code>yfinance</code> , free academic tier
NBER Business Cycle Dating	Official U.S. recession start and end dates	<code>nber.org</code> , public
University of Michigan	Monthly consumer-survey-based inflation expectations and sentiment	FRED + <code>umich.edu</code> , free

Table 2: Validation data sources for outcome correlation analysis.

7 Embedding Strategy and Look-Ahead Bias

7.1 The Concern

Using a 2024-trained embedding model to embed 2015 articles introduces a form of look-ahead bias: the embedding model implicitly “knows” what events occurred between 2015 and 2024, which can subtly position historical articles in semantic space using information unavailable at the time of writing. For a project that emphasizes methodological rigor, this concern must be addressed honestly.

7.2 Our Approach

Embedding choice: `all-mpnet-base-v2`, a general-purpose sentence transformer trained on diverse non-finance-specific data with a cutoff around 2020–2021. This is preferred over finance-specialized models like `FinTextSim`, which are fine-tuned on recent financial text and would amplify look-ahead concerns precisely on our domain of study.

Tradeoff acknowledged. A general-purpose embedding produces marginally noisier financial clusters than a finance-tuned alternative. We accept this tradeoff because look-ahead-bias contamination is more damaging to validity than slightly noisier clusters are to interpretability.

Sensitivity check (mandatory). We compare cluster quality and stability metrics across pre-2021 and post-2021 sub-periods. If clusters are dramatically cleaner in the post-cutoff period, we have evidence of significant look-ahead and must caveat further. If quality is comparable, we have evidence the bias is bounded.

Robustness check (recommended¹). Where feasible, we re-run validation on early anchor narratives using an earlier-trained sentence transformer (e.g., `bert-base-nli-stsb-mean-tokens`, training cutoff ~2018) to verify that our results do not depend on the post-cutoff knowledge of the primary embedding model.

Live operation is methodologically clean. For weekly updates going forward, the embedding model has no future information about newly-arriving articles. Look-ahead is purely a historical-validation concern.

7.3 Why This Does Not Undermine the Project

The project’s contribution is the framework—the combination of narrative detection, epidemiological modeling, descriptive framing, and live deployment—not the embedding model. The framework would work equally well with future, less-contaminated embeddings; the embedding is implementation detail. Acknowledging the bound on look-ahead bias and explicitly evaluating its magnitude is the honest, methodologically defensible path.

8 Methodological Rigor and Risk Mitigations

The project’s credibility rests on a stack of pre-committed disciplines. This section enumerates each risk and the corresponding mitigation.

¹[[Compress](#)]: this earlier-embedding robustness check can be cut for a compressed timeline without seriously weakening the project.

8.1 Narrative Identity Risk

Threat. Narrative boundaries are philosophically slippery. When does a “new” narrative begin versus continue an old one? Different choices produce different SIR fits and downstream measurements.

Mitigations.

- **Algorithmic boundaries.** Narratives are defined operationally by clustering output with pre-specified parameters. We do not impose human judgments about narrative starts and ends.
- **Sensitivity analysis.** Three pre-specified parameter settings (strict, default, permissive) are run in parallel; conclusions must hold across all three.
- **Hierarchical structure.** Three granularity levels allow analysis at multiple scales, accommodating the fact that narratives can be “new” at fine grain and “continuing” at coarse grain.
- **Anchor-narrative validation.** Operational boundaries are validated against documented historical timing for ten anchor narratives.

8.2 Cluster Stability Risk

Threat. Lexical drift in financial language and BERTopic’s stochastic components can produce unstable clusters across runs, undermining downstream analysis.

Mitigations.

- **Bootstrap stability evaluation.** Twenty bootstrap replicates with deterministic seeds; report Normalized Mutual Information (NMI) and Adjusted Rand Index (ARI) across all pairs.
- **Hierarchical merging.** Fine clusters that drift can be tracked at parent levels, providing stability where granular tracking fails.
- **Manual validation.** Ten known anchor narratives are validated by hand to confirm clustering produces semantically reasonable groupings.

8.3 Reverse Causality Risk

Threat. Markets often move before extensive narrative coverage emerges. Apparent “narrative-then-volatility” patterns may reflect news lag rather than narrative causation.

Mitigations.

- **Descriptive reframing.** The project’s primary contribution is measurement, not prediction. Predictive analysis is footnoted only.
- **Source filtering toward analytical content.** The whitelist favors analytical and policy outlets over real-time trading wires, structurally reducing the reverse-causality concern at the data level.
- **Granger causality testing.**² For exploratory predictive analysis, run tests in both directions and report both.
- **Honest acknowledgment.** The paper acknowledges that perfect causal identification is not possible without randomized experiments.

²[Compress]: full Granger testing in both directions is recommended but compressible.

8.4 Multiple-Testing and False Discovery Risk

Threat. Testing many narratives $\times$ many horizons $\times$ many outcomes generates spurious significance by chance.

Mitigations.

- **Pre-registration.** A public document specifies all hypotheses, variables, statistical tests, and decision criteria *before* examining outcome data. Committed to a timestamped GitHub or OSF registration.
- **Walk-forward validation.** Strict temporal split (training: 2010–2019; testing: 2020–present) with no peeking.
- **Benjamini-Hochberg FDR correction** applied across all reported hypotheses.
- **Theoretical pre-justification.** Each tested association requires an articulated economic rationale.

8.5 Survivorship Bias Risk

Threat. Validation against well-known anchor narratives like SVB or COVID is biased toward narratives that survived to become famous, telling us nothing about how the system handles narratives that emerge and fizzle.

Mitigations.

- **Fizzled-narrative validation set.** For each significant anchor narrative, identify a contemporaneous narrative that emerged and dissipated without becoming significant. Validate that the system produces appropriately different stage classifications.
- **Ex-ante significance threshold.** Define narrative significance algorithmically (e.g., articles per day exceeding X for Y consecutive days) before examining historical outcomes.
- **Base-rate reporting.** Report what fraction of detected emergences become significant.

8.6 Data Quality Risk

Threat. GDELT has documented quality issues: approximately 55% accuracy on key fields, ~20% redundancy, Western/U.S. media overrepresentation.

Mitigations.

- **Curated whitelist.** We do not ingest GDELT firehose; we use it only to discover articles from our pre-specified high-quality outlet list.
- **Aggressive deduplication.** MinHash near-duplicate detection within rolling windows.
- **Cross-source robustness check.**³ Validation on a second data source (Alpha Vantage news API) for recent data confirms results are not GDELT-specific artifacts.

³[Compress]: secondary-source robustness check can be cut for compressed timelines.

8.7 Reproducibility Risk

Threat. Future researchers cannot replicate findings if dependencies, data snapshots, or random seeds are not pinned.

Mitigations. Pinned versions for all dependencies (`requirements.txt`); pipeline configuration in version-controlled YAML; archived data snapshot for the historical analysis; deterministic random seeds throughout; public GitHub repository.

9 Kill Criteria

Pre-committed thresholds at which the project pauses, pivots, or accepts graceful degradation. These protect against sunk-cost bias and force honest mid-project assessment.

Criterion	Threshold	Check Date	Response if Hit
Cluster stability	Bootstrap NMI < 0.40 across all parameter settings	End of Phase 1 (pilot)	Investigate; if unsalvageable, narrow to inflation-only fallback scope
Anchor recovery	Fewer than 7 of 10 anchor narratives recovered within 14-day tolerance	End of Phase 3	Debug embedding/filtering; if persistent, shift to novelty-velocity framework
Dynamics fit quality	Median $R^2 < 0.30$ across validation narratives, or R_0 posterior CIs spanning > 2 units	End of Phase 3	Drop SIR; fall back to logistic growth or non-parametric raw curve features
Predictive null	No association reaches significance after FDR correction	Phase 4	<i>Not a project killer</i> —report null result honestly as exploratory finding
Live update reliability	$> 25\%$ of weeks fail update during testing period	Live operation month 1	Degrade to monthly updates; ensure “last updated” is prominent
Time budget	MVP not complete by month 3	Mid-project	Cut from stretch downward; preserve MVP completion above all else

Table 3: Kill criteria with check dates and response plans.

9.1 Fallback Scope (Insurance Policy)

If multiple kill criteria signal stress during the pilot, the pre-committed fallback is to narrow scope to **inflation discourse only, 2018–present**. This reduces corpus size by approximately half, increases semantic homogeneity, shortens the look-ahead concern window, and provides cleaner validation against CPI/PCE data. The methodology, dashboard, and tooling all transfer; only the corpus changes.

10 Validation Strategy

10.1 Anchor Narratives

Ten pre-specified historical narratives serve as the primary validation set. The system must recover their emergence within a 14-day tolerance to be considered functioning.

Narrative	Reference Date	Why It Anchors
SVB collapse	2023-03-09	Entity-rich, sharp emergence, well-documented
GameStop short squeeze	2021-01-22	Distinctive vocabulary, abrupt onset
COVID market crash	2020-02-24	Largest single-narrative event in window
Brexit aftermath	2016-06-24	Long-tail narrative with clear ignition
FTX collapse	2022-11-07	Entity-rich, fast emergence
Transitory inflation debate	2021-Q2	Slower, more diffuse but well-documented
Credit Suisse stress	2023-03-15	Adjacent to SVB; tests cluster differentiation
Regional banking contagion	2023-03-13	Tests narrative branching from anchor events
2022 inflation peak narrative	2022-Q2/Q3	Multi-month evolving narrative
Soft landing emergence	2023-Q3/Q4	Recent, slow emergence, uncontested timing

Table 4: Anchor narratives for validation.

10.2 Fizzled Counterparts

For each anchor, a contemporaneous narrative that emerged but did not become significant. Examples: First Republic concerns adjacent to SVB; intermittent recession-watch narratives that did not crystallize. The fizzled-narrative set is identified before validation runs and is not modified afterward.

10.3 Out-of-Sample Discipline

Training data: 2010–2019. Held-out validation period: 2020–present. The held-out period is not examined until final analysis. All hyperparameters, clustering thresholds, and methodological choices are locked before any held-out evaluation.

10.4 Statistical Reporting Standards

All quantitative claims report 95% bootstrap confidence intervals. All multiple-comparison contexts report both raw p -values and Benjamini-Hochberg-corrected significance. Any predictive analysis additionally reports the Deflated Sharpe Ratio if portfolio-style framings are used.

11 Dashboard Design

11.1 Two Primary Views

View 1: Life-Cycle Viewer. A user selects a narrative (current or historical) and sees its growth curve over time, with the fitted dynamics model overlaid, the estimated R_0 displayed (with credible interval), the current life-cycle stage labeled, representative headlines from key moments annotated, and links to original source articles.

View 2: Emerging Narratives Panel. The forward-looking view. Three sections:

- *Currently emerging.* Narratives detected as crossing from pre-emergence into early-spread within the past 7–30 days. Each shows a brief auto-generated description, R_0 estimate with credible interval, current article volume, and sample headlines.
- *Historical analogs.* For each emerging narrative, the top 3–5 historical narratives with most similar early-stage trajectories.

- *Aggregate state.* Summary metrics characterizing the current discourse landscape (number of narratives in each stage, dominant clusters, etc.).

Stretch view⁴: Narrative Map. A 2D UMAP projection of recent articles with cluster centroids labeled, allowing zoom and click-to-explore.

11.2 Empty-State Handling

When no narratives are detected as freshly emerging, the Emerging Panel falls back to displaying the highest-velocity narratives currently in any stage, with explanatory text. The dashboard never displays as empty.

11.3 Onboarding

A persistent “what is this?” page explains the framework in plain language, with a worked example of a historical narrative’s life-cycle. Tooltips on technical terms (R_0 , stage labels, etc.) make the dashboard accessible without specialist knowledge.

11.4 Live Updating Architecture

Updates run weekly via a cron job on UChicago RCC compute or a small VPS. The job pulls the past week’s articles, embeds them, assigns to existing clusters or flags as candidate new clusters, refits parameters on changed clusters, and writes static JSON/HTML artifacts to the hosting layer. The frontend (deployed to Hugging Face Spaces or Vercel free tier) reads only static artifacts—no live computation occurs at user request time.

If a weekly update fails, the dashboard displays the last-good state with a prominent “last updated” timestamp. Reliability degradation is visible rather than hidden.

12 Build Plan

The project is staged into seven phases. For each phase, items marked [MVP] are essential and cannot be cut; items marked [Compress] are compressible without core damage; items marked [Stretch] are pure additions.

12.1 Phase 1: Pilot and De-Risking

Objective. Validate that the methodology can work on a small sample before committing to the full project.

- [MVP] Pull six months of GDELT data filtered to the curated whitelist.
- [MVP] Set up `all-mpnet-base-v2` embedding pipeline.
- [MVP] Run BERTopic on the pilot corpus; verify clusters are interpretable.
- [MVP] Validate detection of three known recent anchor narratives.
- [MVP] Bootstrap stability check on pilot clusters.
- [Compress] Pre-test SIR fitting on the recovered anchor narratives.

⁴[MVP]: the map view is a stretch feature; the project is complete with the two primary views.

Decision gate. If kill criteria 1 or 2 trigger here, narrow to fallback scope or pivot methodology. If pilot signals are clean, commit to full project.

12.2 Phase 2: Data Pipeline

- [MVP] Confirm UChicago library access for full-text retrieval.
- [MVP] Build full GDELT ingestion for 2010–present, filtered to whitelist.
- [MVP] Build Federal Reserve, IMF, BIS, OECD, NBER ingestion for the same period.
- [MVP] MinHash deduplication.
- [MVP] Topic filtering pipeline.
- [MVP] Storage layer with checkpointing.
- [Compress] Cross-source robustness check infrastructure (Alpha Vantage news API).

12.3 Phase 3: Embedding, Clustering, and Dynamics

- [MVP] Embed full historical corpus with `all-mpnet-base-v2`.
- [MVP] BERTopic clustering with three granularity levels.
- [MVP] Bootstrap stability evaluation reporting NMI and ARI.
- [MVP] Look-ahead sensitivity check (cluster quality by sub-period).
- [MVP] Logistic growth fitting for all clusters (the most stable model; suitable as MVP fallback).
- [Compress] Multi-model fitting (SIR, Gompertz, exponential) with Bayesian inference.
- [Compress] Earlier-embedding robustness check.
- [Stretch] Online incremental clustering for new-narrative detection (vs. batch re-clustering).

12.4 Phase 4: Validation

- [MVP] Pre-registration document committed to public timestamp before any predictive analysis.
- [MVP] Anchor narrative recovery on full 10-narrative set.
- [MVP] Fizzled-narrative validation.
- [MVP] Sensitivity analysis across three parameter settings.
- [Compress] Exploratory predictive analysis with Granger causality and FDR-corrected significance.

12.5 Phase 5: Dashboard Build

- [MVP] Streamlit or similar single-view dashboard (Life-Cycle Viewer).
- [MVP] Static cached artifacts produced by analysis pipeline.
- [MVP] Onboarding “what is this?” page with worked example.
- [MVP] Public deployment (Hugging Face Spaces).

- [Compress] Emerging Narratives Panel (full version with historical analogs).
- [Stretch] Narrative Map view with UMAP projection.

12.6 Phase 6: Live Updating

- [Compress] Weekly cron-based update pipeline.
- [Compress] Incremental embedding and cluster assignment for new articles.
- [Compress] Robust failure handling and “last updated” display.
- [Stretch] Notification system for newly-detected emerging narratives.

Live Updating Note

The live updating system is genuinely valuable but is the single most cuttable component if time pressures escalate. The MVP delivers a static snapshot; the full project delivers weekly extensions. Both produce a credible deployed artifact. The choice should be driven by time available, not by ambition.

12.7 Phase 7: Writeup and Reproducibility

- [MVP] GitHub repository with pinned dependencies, clear README, and replication instructions.
- [MVP] Short technical report (5–8 pages) describing methodology and findings.
- [Compress] Workshop-paper-quality writeup (12–15 pages).
- [Stretch] Submission to a workshop venue (e.g., ACL workshop on Economics and NLP, NeurIPS workshop on ML for Finance).

12.8 Speedrun MVP Summary

If circumstances require the fastest possible build, the speedrun version cuts everything not marked [MVP]. This produces:

Speedrun MVP deliverable: static historical analysis 2010–present on macro discourse, with logistic-growth life-cycle fitting on hierarchically-clustered narratives, validated on 10 anchor narratives with sensitivity analysis, presented through a single-view public dashboard with onboarding, fully reproducible from a public GitHub repository, accompanied by a 5-8-page technical report. No live updating, no SIR-specific contribution, no exploratory predictive analysis. This is sufficient for a credible undergraduate research artifact. It is missing the live-deployment differentiator and the dynamics-modeling novel contribution, but the core measurement framework is intact.

13 Deliverables

13.1 Primary Deliverables

1. **Public dashboard.** Deployed at a stable URL on free-tier hosting. Two views (Life-Cycle Viewer + Emerging Narratives Panel). Updated weekly. Anyone with the URL can use it

without authentication.

2. **Public GitHub repository.** Full source code, configuration files, pre-registration document, replication scripts, anchor-narrative ground truth, validation metrics. Pinned dependencies via `requirements.txt`. README with installation and replication instructions.
3. **Pre-registration document.** Committed to public timestamp before any held-out validation. Specifies hypotheses, variables, statistical tests, and decision criteria.
4. **Technical report.** Minimum 5–8 pages describing methodology, results, and limitations. Stretch goal: 12–15-page workshop-paper-quality version with submission to a relevant venue.

13.2 Secondary Deliverables

1. **Anchor narrative validation set.** Released as a labeled CSV/JSONL with documented timing, source citations, and significance threshold. Useful infrastructure for future researchers.
2. **Stability and sensitivity analysis report.** Detailed methodological appendix documenting cluster stability across bootstrap replicates and across parameter settings.

14 Resume and Application Signal

When summarized for graduate school applications or interviews, the project produces the following claim: *a publicly-deployed quantitative measurement framework for narrative dynamics in U.S. financial discourse, applying transformer-based semantic clustering and epidemiological growth models to identify and characterize narrative life-cycles, with rigorous methodology including pre-registration, bootstrap stability analysis, and out-of-sample validation, accompanied by a deployed live-updating dashboard and a public technical report.*

The intellectual through-line—LLM reliability → temporal reasoning in finance → narrative dynamics in financial discourse—demonstrates a coherent research arc from initial interest to focused contribution. The combination of NLP methodology, time-series modeling, statistical rigor, and full-stack deployment demonstrates breadth across multiple skill axes. The descriptive framing demonstrates research maturity and intellectual honesty about what can be claimed.

A Pre-Specified Parameters

This appendix lists key parameter choices that must be locked in advance of analysis to support reproducibility and guard against post-hoc tuning.

- **Embedding model:** `all-mpnet-base-v2`, version pinned via `sentence-transformers` library.
- **Article truncation:** headline + first 600 tokens of body.
- **Topic filter keyword list:** pre-specified, version-controlled in repository.
- **UMAP parameters:** `n_neighbors=15`, `min_dist=0.1`, `n_components=5`, `metric='cosine'`.
- **HDBSCAN parameters:** `min_cluster_size=20`, `min_samples=5`, `cluster_selection_method='eom'` (default setting; sensitivity values for parameter sweep: `min_cluster_size ∈ {10, 20, 40}`).
- **Granularity hierarchy:** fine, medium, coarse defined by post-hoc cluster merging at three pre-specified silhouette thresholds.

- **Smoothing window for time series:** 7-day centered moving average.
- **Stage thresholds:** pre-emergence requires fewer than 50 cumulative articles; early-spread requires $R_0 > 1.0$ with peak day not yet reached.
- **Bootstrap replicates:** 20.
- **Anchor narrative tolerance:** 14 days from documented historical date.
- **Random seeds:** pinned in configuration; specific values in `config.yaml`.

B Glossary

- **R_0 (basic reproduction number):** expected number of secondary articles a single narrative-carrying article generates in a fully susceptible discourse population. $R_0 > 1$ indicates growth; $R_0 < 1$ indicates decay.
- **BERTopic:** neural topic modeling framework combining transformer embeddings, UMAP dimensionality reduction, HDBSCAN clustering, and class-based TF-IDF representation (Grootendorst, 2022).
- **Conformal prediction:** distribution-free framework for constructing prediction intervals with finite-sample coverage guarantees.
- **HDBSCAN:** density-based hierarchical clustering algorithm.
- **Look-ahead bias:** contamination from using information unavailable at the time being studied; in our context, embedding models trained on data including events from the analysis period.
- **NMI (Normalized Mutual Information):** cluster stability metric; values range $[0, 1]$ with higher indicating greater agreement.
- **SIR model:** compartmental epidemic model partitioning population into Susceptible, Infected, Recovered groups with first-order ODE dynamics.
- **UMAP:** non-linear dimensionality reduction technique well-suited to visualization and density-based clustering.

References

- Grootendorst, M. (2022). BERTopic: Neural topic modeling with a class-based TF-IDF procedure. *arXiv preprint arXiv:2203.05794*.
- Kermack, W. O. and McKendrick, A. G. (1927). A contribution to the mathematical theory of epidemics. *Proceedings of the Royal Society of London A*, 115(772), 700–721.
- Shiller, R. J. (2017). Narrative economics. *American Economic Review*, 107(4), 967–1004.
- Shiller, R. J. (2019). *Narrative Economics: How Stories Go Viral and Drive Major Economic Events*. Princeton University Press.
- Vitanov, N. K., Vitanov, Z. I., and Kantz, H. (2024). On the mathematical theory of news waves. *Entropy*, 26(1), 5.